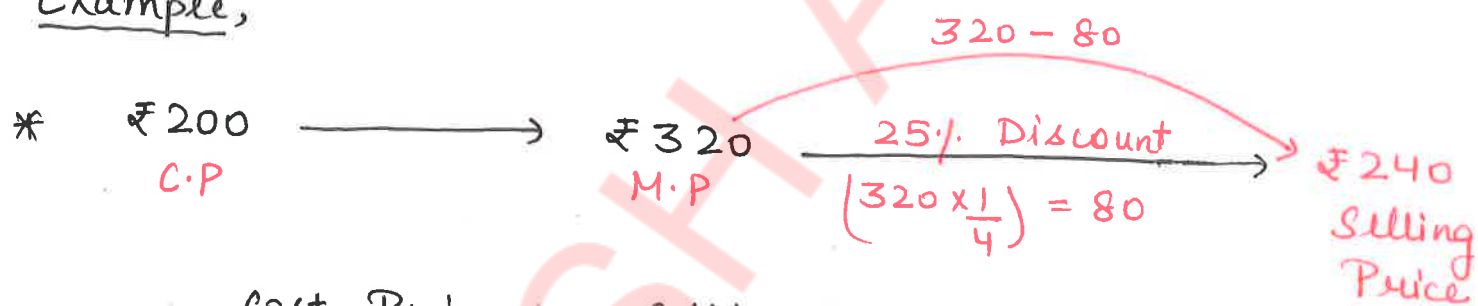


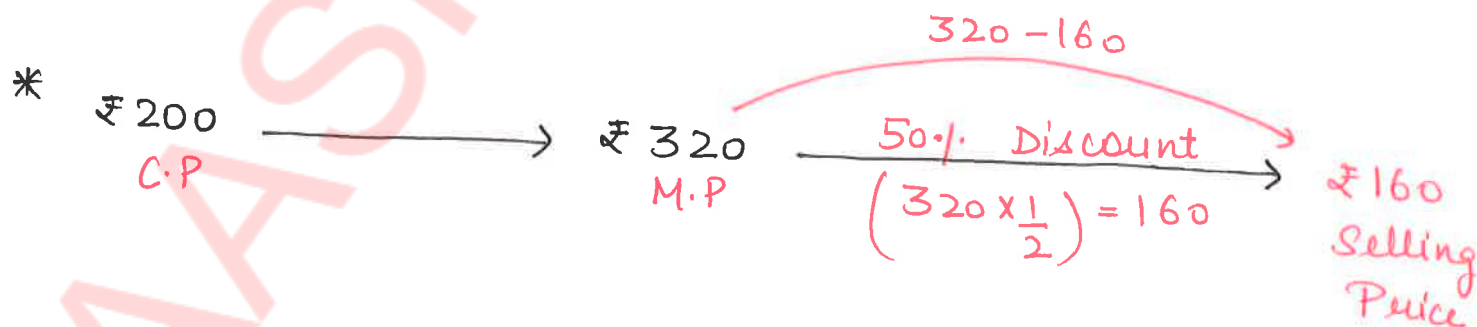
Profit, Loss and Discount :-

- 1) Cost Price : Price at which shopkeeper purchased an item.
{क्रय मूल्य}
- 2) Marked Price : Price marked on the item.
like, Price Tag.
{अंकित मूल्य}
- 3) Selling Price : Money actually paid by the customer to the shopkeeper.
{विक्रय मूल्य}

Example,



Cost Price Selling Price
200 240
 $\xrightarrow{+ ₹40 \text{ (Profit)}}$



Cost Price Selling Price
200 160
 $\xrightarrow{- ₹40 \text{ (Loss)}}$

⇒ Let's understand the terms in detail :-

1) Mark-up :-

$$\begin{array}{ccc} \text{₹}200 & \xrightarrow{+120} & \text{₹}320 \\ \text{C.P} & & \text{M.P} \end{array}$$

* Item is marked up by ₹120.

$$\text{MP} - \text{CP} = \text{Mark-up}$$

2) Mark-up Percent :-

$$\begin{array}{ccc} \text{₹}200 & \xrightarrow{+120} & \text{₹}320 \\ \text{C.P} & (\text{Mark-up}) & \text{M.P} \end{array}$$

* Markup → ₹120

* Markup Percent → $\frac{\text{₹}120}{\text{₹}200} \times 100 \Rightarrow 60\%$

$$\text{Mark-up \%} = \frac{\text{Mark-up}}{\text{Cost Price}} \times 100$$

3) Discount :-

$$\begin{array}{ccc} \text{₹}320 & \xrightarrow{-80} & \text{₹}240 \\ \text{M.P} & (\text{Discount}) & \text{S.P} \end{array}$$

* Discount = Marked Price - Selling Price

4) Discount Percent :-

$$\begin{array}{ccc} \text{₹ } 320 & \xrightarrow{-80} & \text{₹ } 240 \\ \text{M.P} & & \text{S.P} \end{array}$$

* Discount $\rightarrow$ ₹ 80

* Discount Percent $\rightarrow \frac{\text{₹ } 80}{\text{₹ } 320} \times 100 = 25\%$

$$\text{Discount \%} = \frac{\text{Discount}}{\text{Marked Price}} \times 100$$

5) Profit :-

$$\begin{array}{ccc} \text{₹ } 200 & \xrightarrow{+40} & \text{₹ } 240 \\ \text{C.P} & \text{(Profit)} & \text{S.P} \end{array}$$

* Profit $\rightarrow$ ₹ 40

• In case of Profit,
 $CP < SP$

6) Profit Percent :-

$$\begin{array}{ccc} \text{₹ } 200 & \xrightarrow{+40} & \text{₹ } 240 \\ \text{C.P} & & \text{S.P} \end{array}$$

* Profit = ₹ 40

* Profit Percent = $\frac{40}{200} \times 100 = 20\%$

$$\text{Profit \%} = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$

7) Loss :-

$$\begin{array}{ccc} \text{₹ 200} & \xrightarrow{-40} & \text{₹ 160} \\ \text{C.P} & & \text{S.P} \end{array}$$

* Loss $\rightarrow$ ₹ 40

• In case of Loss,
 $CP > SP$.

8) Loss Percent :-

$$\begin{array}{ccc} \text{₹ 200} & \xrightarrow{-40} & \text{₹ 160} \\ \text{C.P} & & \text{S.P} \end{array}$$

* Loss $\rightarrow$ ₹ 40

* Loss Percent $\rightarrow \frac{\text{₹ 40}}{\text{₹ 200}} \times 100 = 20\%$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{Cost Price}} \times 100$$

9) No Profit, No Loss.

$$\begin{array}{ccc} \text{₹ 200} & \xrightarrow{+0} & \text{₹ 200} \\ \text{C.P} & & \text{S.P} \end{array}$$

* When the item is sold at the same price it was purchased by the shopkeeper there is No Profit or No Loss.

• In this case, $CP = SP$.

Percentages

- Discount Percent
- Loss Percent
- Profit Percent
- Mark Up Percent

Base Values

Marked up Price

Cost Price

Cost Price

Cost Price

When No Discount is given,

$$\begin{array}{ccccc} \text{₹ } 200 & \xrightarrow{\substack{+120 \\ \text{Mark-up}}} & \text{₹ } 320 & \xrightarrow{\substack{\text{No} \\ \text{Dis.}}} & \text{₹ } 320 \\ \text{CP} & & \text{MP} & & \text{SP} \end{array}$$

$$\begin{array}{ccc} \text{CP} & \xrightarrow{\substack{+120 \\ \text{Mark-up}}} & \text{MP} \\ 200 & & 320 \end{array}$$

$$\begin{array}{ccc} \text{CP} & \xrightarrow{\substack{+120 \\ \text{Profit}}} & \text{SP} \\ 200 & & 320 \end{array}$$

$$\text{MP} = \text{SP}$$

$$\text{Markup} = \text{Profit}$$

$$\text{Markup \%} = \text{Profit \%}$$

⇒ Example,

$$\begin{array}{ccccc} \text{CP} & \xrightarrow{+120} & \text{MP} & \xrightarrow{-80} & \text{SP} \\ 200 & & 320 & & 240 \end{array}$$

+40 Profit (from CP to SP)

$$\begin{array}{ccccccc} \# & \text{Markup} & = & \text{Profit} & + & \text{Discount} \\ & \downarrow & & \downarrow & & \downarrow \\ & 120 & & 40 & & 80 \end{array}$$

{* Do not consider signs}

II)



* When you give more discount than Markup,

$$\begin{array}{ccccc} \text{Discount} & - & \text{Markup} & = & \text{Loss} \\ \downarrow & & \downarrow & & \downarrow \\ 250 & & 200 & & 50 \end{array}$$

{ * Do not consider signs.
Consider only values }

In case of Profit,

$$* \text{ Discount} + \text{Profit} = \text{Markup}$$

In case of Loss,

$$* \text{ Discount} - \text{Loss} = \text{Markup}$$

{ with only values
and no sign }

Q1: A book was bought for Rs. 164 and sold for Rs. 213. Find the profit.
 एक किताब 164 रुपये में खरीदी गई और 213 रुपये में बेची गई। लाभ ज्ञात कीजिए।

(A) Rs. 45

(B) Rs. 49

(C) Rs. 53

(D) Rs. 57

(E) Rs. 61

Cost Price

₹ 164

Selling Price

₹ 213

+ ₹ 49

(Profit)

{ CP < SP }

Q2: A pen stand was bought for Rs. 137 and sold for Rs. 109. Find the loss.
 एक पेन स्टैंड 137 रुपये में खरीदा गया और 109 रुपये में बेचा गया। हानि ज्ञात कीजिए।

(A) Rs. 22

(B) Rs. 24

(C) Rs. 26

(D) Rs. 28

(E) Rs. 30

Cost Price

₹ 137

Selling Price

₹ 109

₹ 28

(Loss)

{ CP > SP }

Q3: A chair was sold for Rs. 672 at a profit of Rs. 84. Find the cost price.
 एक कुर्सी 672 रुपये में बेची गई, जिस पर 84 रुपये का लाभ हुआ। लागत मूल्य ज्ञात कीजिए।

(A) Rs. 580

(B) Rs. 588

(C) Rs. 596

(D) Rs. 604

(E) Rs. 612

Cost Price

₹ 588

Selling Price

₹ 672

+ ₹ 84

profit

$$\left\{ \begin{array}{l} 672 - 84 \\ = ₹ 588 \end{array} \right\}$$

Q4: A bottle was bought for Rs. 348 and sold at a profit of Rs. 57. Find the selling price.
 एक बोतल 348 रुपये में खरीदी गई और 57 रुपये के लाभ पर बेची गई। विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 397

(B) Rs. 401

☒ (C) Rs. 405

(D) Rs. 409

(E) Rs. 413

Cost Price

₹ 348

Selling Price

₹ 405

+ ₹57
(Profit)

$$\{ 348 + 57 \} = 405$$

Q5: A box was sold for Rs. 436 at a loss of Rs. 39. Find the cost price.
 एक डिब्बा 436 रुपये में बेचा गया, जिस पर 39 रुपये का नुकसान हुआ। लागत मूल्य ज्ञात कीजिए।

☒ (A) Rs. 475

(B) Rs. 485

(C) Rs. 495

(D) Rs. 505

(E) Rs. 515

Cost Price

₹ 475

Selling Price

₹ 436

- ₹39 (Loss)

$$SP + Loss = CP$$

$$436 + 39 = ₹ 475$$

Q6: A toy was bought for Rs. 529 and sold at a loss of Rs. 46. Find the selling price.
 एक खिलौना 529 रुपये में खरीदा गया और 46 रुपये की हानि पर बेचा गया। विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 459

(B) Rs. 465

(C) Rs. 471

(D) Rs. 477

☒ (E) Rs. 483

Cost Price

₹ 529

Selling Price

₹ 483

- ₹46
Loss

$$CP - Loss = SP$$

$$529 - 46 = 483$$

Q7: An article has a marked price of Rs. 684 and a discount of Rs. 73. Find the selling price.

एक वस्तु का अंकित मूल्य 684 रुपये है और उस पर 73 रुपये की छूट है। विक्रय मूल्य ज्ञात कीजिए।

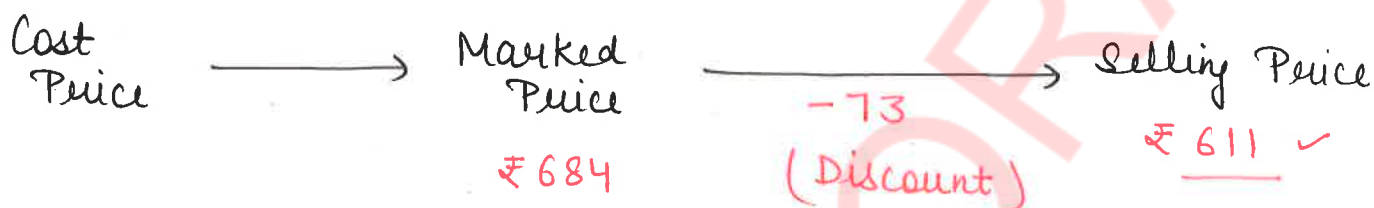
(A) Rs. 599

(B) Rs. 603

(C) Rs. 607

☒ (D) Rs. 611

(E) Rs. 615



$$\begin{aligned} \text{MP} - \text{Discount} &= \text{SP} \\ 684 - 73 &= 611 \end{aligned}$$

Q8: An article has a marked price of Rs. 932 and is sold for Rs. 867. Find the discount.

एक वस्तु का अंकित मूल्य 932 रुपये है और वह 867 रुपये में बेची गई। छूट ज्ञात कीजिए।

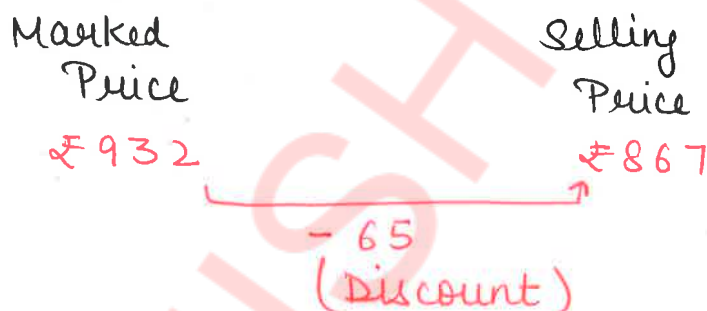
(A) Rs. 60

☒ (B) Rs. 65

(C) Rs. 70

(D) Rs. 75

(E) Rs. 80



$$\begin{aligned} * \text{ MP} - \text{SP} &= \text{Discount} \\ 932 - 867 &= 65 \end{aligned}$$

Q9: An item is sold for Rs. 823 after a discount of Rs. 87. Find the marked price.

एक वस्तु 87 रुपये की छूट के बाद 823 रुपये में बेची जाती है। अंकित मूल्य ज्ञात कीजिए।

(A) Rs. 898

(B) Rs. 904

(C) Rs. 922

(D) Rs. 916

☒ (E) Rs. 910



$$\begin{aligned} \text{MP} - \text{Discount} &= \text{SP} \\ \text{MP} - 87 &= 823 \\ \text{MP} &= 910 \end{aligned}$$

Q10: An article costs Rs. 250 and is sold for Rs. 315. Find the profit percentage.

एक वस्तु की लागत 250 रुपये है और उसे 315 रुपये में बेचा जाता है। लाभ प्रतिशत ज्ञात कीजिए।

(A) 18%

(B) 20%

(C) 22%

(D) 24%

(E) 26%

$$\begin{array}{ccc} \text{CP} & & \text{SP} \\ 250 & & 315 \\ \hline & +65 & \\ & \text{Profit} & \end{array}$$

$$\# \text{ Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$$\text{Profit \%} = \frac{65}{250} \times 100 = 26\%$$

Q11: An article costs Rs. 460 and is sold for Rs. 391. Find the loss percentage.

एक वस्तु की लागत 460 रुपये है और उसे 391 रुपये में बेचा गया है। हानि प्रतिशत ज्ञात कीजिए।

(A) 15%

(B) 18%

(C) 21%

(D) 24%

(E) 27%

$$\begin{array}{ccc} \text{CP} & & \text{SP} \\ 460 & & 391 \\ \hline & -69 & \\ & \text{Loss} & \end{array}$$

$$\# \text{ Loss \%} = \frac{\text{Loss}}{\text{CP}} \times 100$$

$$\begin{aligned} \text{Loss \%} &= \frac{69}{460} \times 100 \\ &= 15\% \end{aligned}$$

Q12: An article is sold for Rs. 708 at a profit of 18%. Find the cost price.

एक वस्तु 708 रुपये में 18% लाभ पर बेची जाती है। लागत मूल्य ज्ञात कीजिए।

(A) Rs. 560

(B) Rs. 575

(C) Rs. 540

(D) Rs. 600

(E) Rs. 620

$$\# 18\% = \frac{9}{50}$$

$$\begin{array}{ccc} \text{CP} & : & \text{SP} \\ 50 & : & 59 \\ \downarrow \times 12 & & \downarrow 12 \\ \underline{\underline{600}} & & \underline{\underline{708}} \end{array}$$

$$\text{CP} (100\% + 18\%) = \text{SP}$$

$$\text{CP} \times 118\% = \text{SP}$$

$$\text{CP} \times \frac{118}{100} = 708$$

$$\text{CP} = 708 \times \frac{100}{118}$$

$$\boxed{\text{CP} = 600}$$

Q13: An article costs Rs. 456 and is sold at a profit of $37\frac{1}{2}\%$. Find the selling price.

एक वस्तु की लागत 456 रुपये है और इसे $37\frac{1}{2}\%$ लाभ पर बेचा जाता है। विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 615

(B) Rs. 627

(C) Rs. 639

(D) Rs. 651

(E) Rs. 663

$$\Rightarrow 37\frac{1}{2}\% = 37.5\% = \frac{3}{8}$$

$$\# 456 \times \frac{3}{8} = 171$$

CP		SP
456		627
$\xrightarrow{+171}$		

$$CP (100\% + 37.5\%) = SP$$

$$CP \left(1 + \frac{3}{8}\right) = SP$$

$$CP \times \frac{11}{8} = SP$$

$$456 \times \frac{11}{8} = SP$$

$$\boxed{SP = 627}$$

Q14: An article is sold for Rs. 731 at a loss of 14%. Find the cost price.

एक वस्तु 731 रुपये में 14% की हानि पर बेची गई। उसका लागत मूल्य ज्ञात कीजिए।

(A) Rs. 814

(B) Rs. 910

(C) Rs. 920

(D) Rs. 875

(E) Rs. 850

$$CP \xrightarrow{-14\%} SP$$

$$* 14\% = \frac{7}{50}$$

CP	:	SP
50	:	43

$\times 17$

$$\boxed{850}$$

$$\begin{array}{c} 17 \\ \downarrow \\ \cancel{731} \end{array}$$

$$CP (100\% - 14\%) = SP$$

$$CP \left(1 - \frac{7}{50}\right) = SP$$

$$CP \times \frac{43}{50} = SP$$

$$CP = 731 \times \frac{50}{43}$$

$$CP = 17 \times 50$$

$$\boxed{CP = 850}$$

$$\left\{ 729 \times \frac{1}{9} \Rightarrow 81 \right\}$$

$$\begin{array}{rcl} \text{CP} & & \text{SP} \\ 729 & \xrightarrow{-81} & 648 \end{array}$$

Q15: An article costs Rs. 729 and is sold at a loss of $11 \frac{1}{9}\%$. Find the selling price.

एक वस्तु की लागत 729 रुपये है और इसे $11 \frac{1}{9}\%$ की हानि पर बेचा जाता है। विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 624

(B) Rs. 636

(C) Rs. 648

(D) Rs. 660

(E) Rs. 672

$$* 11 \frac{1}{9} \% = 11.11 \% = \frac{1}{9}$$

$$\begin{array}{ccc} \text{CP} & : & \text{SP} \\ \cancel{9} & : & 8 \\ \downarrow & & \downarrow \times 81 \\ 81 & & \boxed{648} \end{array}$$

$$\text{CP} (100\% - 11.11\%) = \text{SP}$$

$$\text{CP} \left(1 - \frac{1}{9}\right) = \text{SP}$$

$$\text{CP} \times \frac{8}{9} = \text{SP}$$

$$729 \times \frac{8}{9} = \text{SP}$$

$$\boxed{\text{SP} = 648}$$

Q16: An article costs Rs. 575 and its marked price is Rs. 736. Find the markup amount.

एक वस्तु की लागत 575 रुपये है और उसका अंकित मूल्य 736 रुपये है। लाभ की राशि ज्ञात कीजिए।

(A) Rs. 143

(B) Rs. 149

(C) Rs. 155

(D) Rs. 161

(E) Rs. 167

Cost Price

575

Marked Price

736

+ 161
(Markup)

Q17: An article has a marked price of Rs. 700 and is sold for Rs. 581. Find the discount percentage.

एक वस्तु का अंकित मूल्य 700 रुपये है और वह 581 रुपये में बेची जाती है। छूट प्रतिशत ज्ञात कीजिए।

(A) 13%

(B) 15%

(C) 17%

(D) 19%

(E) 21%

$$\begin{array}{ccc} \text{MP} & & \text{SP} \\ 700 & \xrightarrow{-119} & 581 \\ & & \text{(Discount)} \end{array}$$

$$\# \text{ Discount \%} = \frac{\text{Discount}}{\text{MP}} \times 100$$

$$\text{Discount \%} = \frac{119}{700} \times 100$$

$$= 17\%$$

Q18: An article has a marked price of Rs. 1140 and a discount of $16\frac{2}{3}\%$. Find the selling price.

एक वस्तु का अंकित मूल्य 1140 रुपये है और उस पर $16\frac{2}{3}\%$ की छूट है। विक्रय मूल्य ज्ञात कीजिए।

(A) Rs. 950

(B) Rs. 960

(C) Rs. 970

(D) Rs. 870

(E) Rs. 925

$$* 16\frac{2}{3}\% = 16.66\% = \frac{1}{6}$$

MP	SP
1140	950
$\xrightarrow{-190}$	

$$\bullet \left\{ 1140 \times \frac{1}{6} = 190 \right\}$$

MP	SP
1140	1140
$\downarrow$	$\downarrow$
190	$\times 190$
1140	$\boxed{950}$

Q19: An item is sold for Rs. 891 after a discount of $8\frac{1}{3}\%$. Find the marked price.

एक वस्तु $8\frac{1}{3}\%$ की छूट के बाद 891 रुपये में बेची जाती है। अंकित मूल्य ज्ञात कीजिए।

(A) Rs. 924

(B) Rs. 936

(C) Rs. 948

(D) Rs. 960

(E) Rs. 972

$$* 8\frac{1}{3}\% = 8.33\% = \frac{1}{12}$$

MP	SP
972	891
$\xrightarrow{-81}$	

$$\# MP \times \frac{11}{12} = SP$$

$$MP = 891 \times \frac{12}{11}$$

$$MP = 972$$

$$\bullet 8.33\% = \frac{1}{12}$$

MP	SP
12	11
$\downarrow$	$\downarrow$
$\times 81$	$\times 81$
$\boxed{972}$	891

Q20: An article costs Rs. 500 and its marked price is Rs. 635. Find the markup percentage.

एक वस्तु की कीमत 500 रुपये है और उसका अंकित मूल्य 635 रुपये है। लाभ प्रतिशत ज्ञात कीजिए।

(A) 23%

☒ (B) 27%

(C) 31%

(D) 35%

(E) 39%

$$\begin{array}{ccc}
 \text{CP} & & \text{MP} \\
 500 & & 635 \\
 & \xrightarrow{+135} & \\
 & \text{(Markup)} &
 \end{array}$$

$$\begin{aligned}
 \# \text{ Markup } \% &= \frac{\text{Markup}}{\text{CP}} \times 100 \\
 &= \frac{135}{500} \times 100 \\
 &= 27\%
 \end{aligned}$$

Q21: An article costs Rs. 675 and is marked 28% above cost. Find the marked price.

एक वस्तु की कीमत 675 रुपये है और उस पर अंकित मूल्य लागत मूल्य से 28% अधिक है। अंकित मूल्य ज्ञात कीजिए।

(A) Rs. 846

(B) Rs. 855

☒ (C) Rs. 864

(D) Rs. 873

(E) Rs. 882

$$\begin{array}{ccc}
 \text{CP} & & \text{MP} \\
 675 & & 864 \\
 & \xrightarrow{+189} & \\
 & \text{Markup} &
 \end{array}$$

$$\left\{ 675 \times \frac{28}{100} = 189 \right\} \text{ Markup}$$

Q22: The profit on an article is Rs. 118 and the discount on it is Rs. 52. Find MP - CP.

किसी वस्तु पर लाभ 118 रुपये है और उस पर छूट 52 रुपये है। मूल्यहास - क्रय मूल्य ज्ञात कीजिए।

(A) Rs. 152

(B) Rs. 160

(C) Rs. 166

☒ (D) Rs. 170

(E) Rs. 174

$$\begin{aligned}
 \text{Profit} &= ₹118 \\
 \text{Discount} &= ₹52
 \end{aligned}$$

$$\{ \text{MP} - \text{CP} = \text{Markup} \}$$

$$* \text{ Markup} = \text{Discount} + \text{Profit}$$

$$\text{Markup} = 118 + 52$$

$$\text{Markup} = 170$$

$$\text{So, } \text{MP} - \text{CP} = \text{Markup} = \underline{\underline{170}} \checkmark$$